

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

John Roth

Kyle Wagner

Liv d'Aliberti

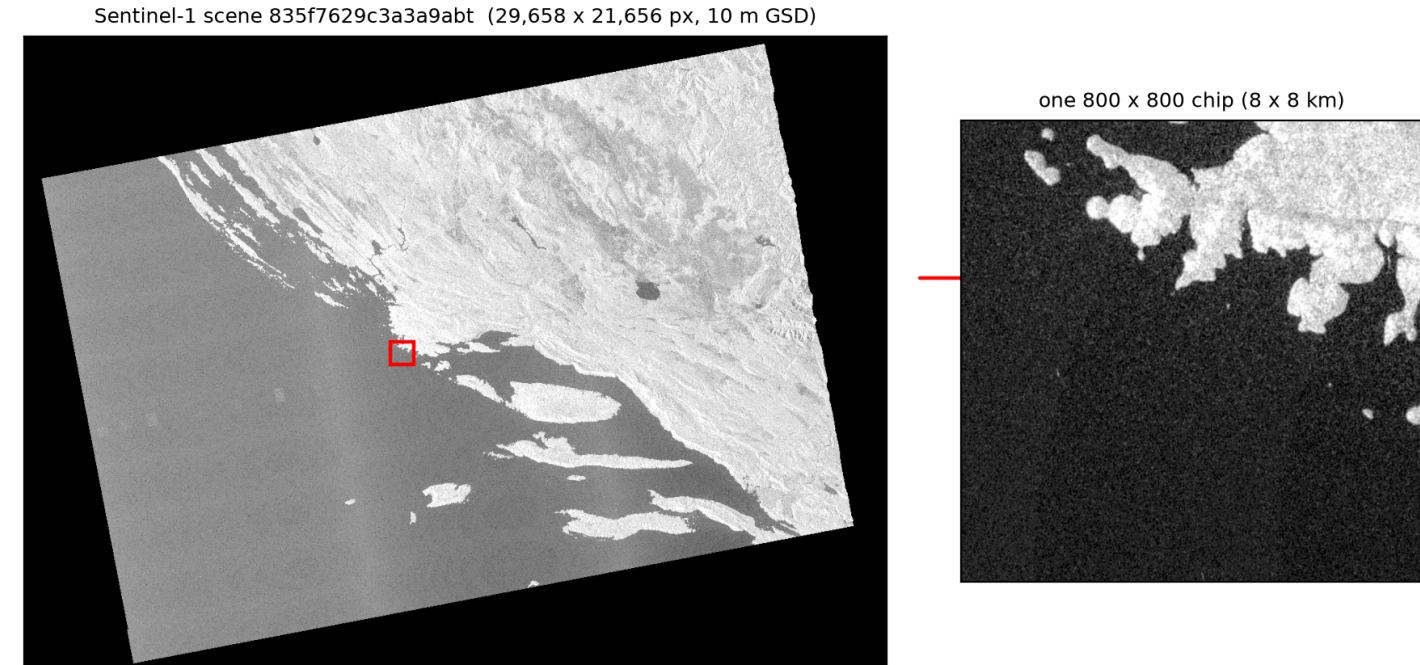
Johns Hopkins University

Short answer: not consistently. With 12 labeled scenes every pretrained encoder beats random initialization, but SAR pretraining beats optical only for the CNN, and only while labels are scarce.

Motivation

- **Dark vessels** have no matching AIS report. Sentinel-1 SAR [2] sees them at night and through cloud.
- Vessels fill a few pixels in a 25,000-pixel scene. Sea clutter and coastlines give strong false returns.
- **Human-verified labels are scarce.** Pretraining can lower the label cost, but which source domain helps most, and does the answer depend on the backbone?

Data: xView3-SAR [1]



One Sentinel-1 scene (about 300 × 220 km) and one 800-pixel chip (8 × 8 km).

- 150 frozen study scenes: **111 train, 23 dev, 16 test**.
- Nested budgets: **12 ⊂ 28 ⊂ 56 ⊂ 111** training scenes (10, 25, 50, 100%).
- **50 near-shore, human-verified scenes**, opened once after test scoring: 8,642 vessels, 3,644 dark, 2,686 within 2 km of shore.
- Input [VH, VV, VH – VV] in dB, 10 m pixels.

References

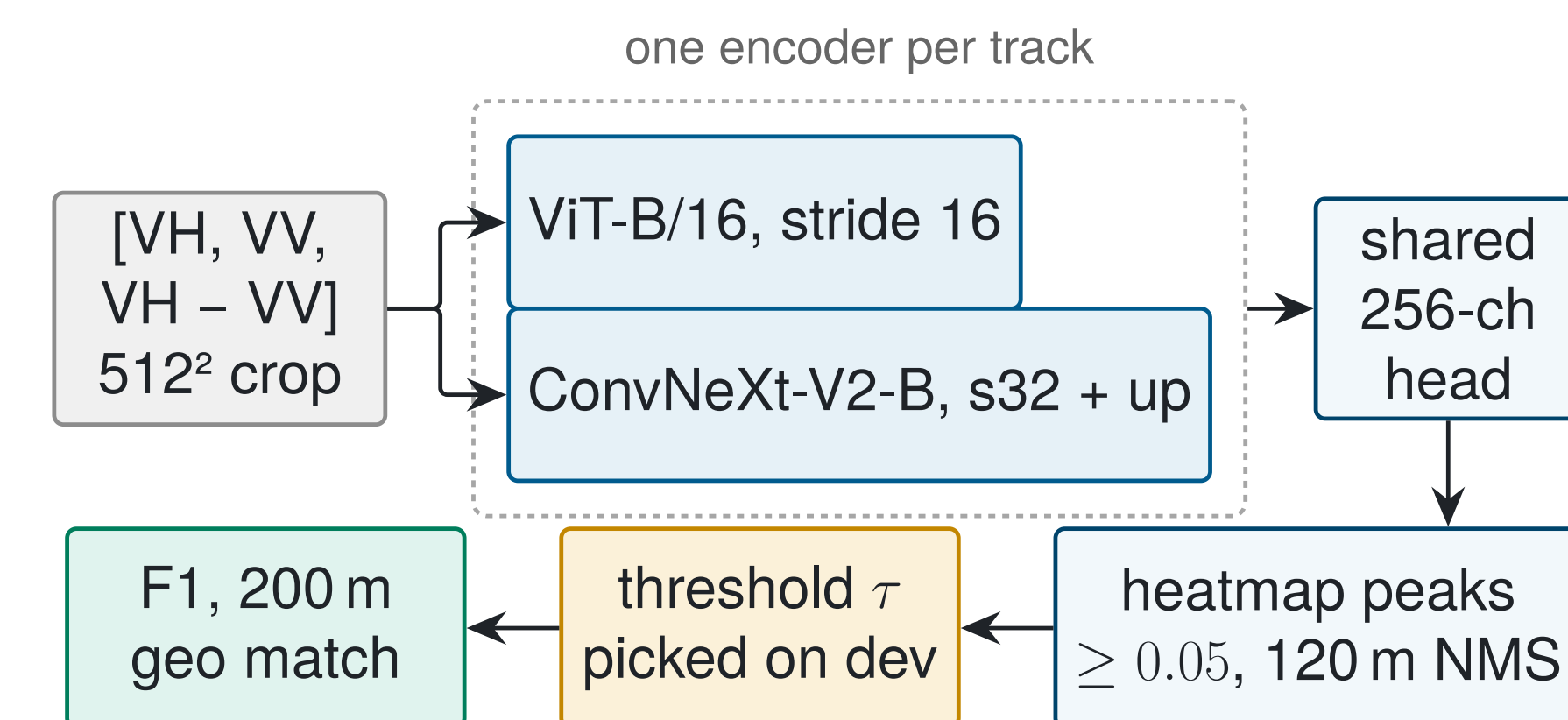
- [1] F. S. Paolo et al. xView3-SAR: Detecting dark fishing activity using SAR imagery. NeurIPS 2022.
- [2] R. Torres et al. GMES Sentinel-1 mission. Remote Sens. Environ. 2012.
- [3] A. Dosovitskiy et al. An image is worth 16 × 16 words. ICLR 2021.
- [4] S. Woo et al. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. CVPR 2023.
- [5] A. Steiner et al. How to train your ViT? TMLR 2022.
- [6] J. Straka and I. Gruber. SatDINO: Self-supervised pretraining for remote sensing. arXiv:2508.21402, 2025.
- [7] D. Liu et al. SARMAE: Masked autoencoder for SAR representation learning. CVPR 2026.
- [8] K. N. Clasen et al. reBEN: Refined BigEarthNet dataset. IGARSS 2025.
- [9] X. Zhou, D. Wang, and P. Krähenbühl. Objects as points. arXiv:1904.07850, 2019.
- [10] T.-Y. Lin et al. Focal loss for dense object detection. ICCV 2017.
- [11] A. Fuller, K. Millard, and J. R. Green. CROMA: Contrastive radar-optical masked autoencoders. NeurIPS 2023.
- [12] Z. Xiong et al. Neural plasticity-inspired multimodal foundation model for Earth observation (DOFA). arXiv:2403.15356, 2024.

8 arms × 4 budgets = 32 cells

Role	ViT-B/16 [3]	ConvNeXt V2 [4]
Random	fresh init	fresh init
Optical	SatDINO [6] (fMoW-RGB)	BigEarthNet-S2 [8]
SAR	SARMAE [7] (SAR-1M)	BigEarthNet-S1 [8]
ImageNet	AugReg [5]	FCMAE [4]

Only the initialization changes inside a track (86M vs. 89M parameters). We compare domains **within** an architecture, then check whether the pattern repeats across the two.

One shared point detector

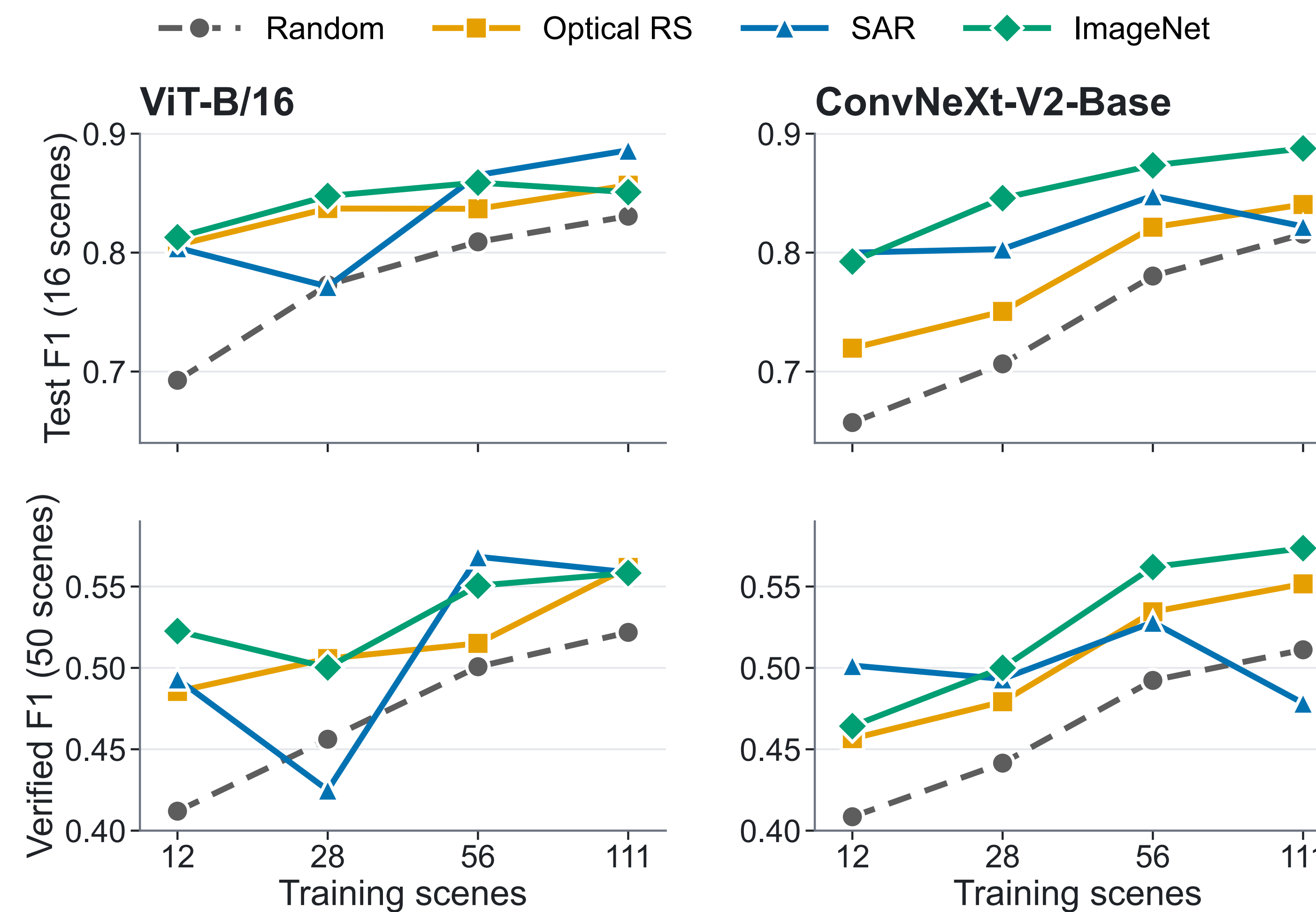


- CenterNet-style heatmap [9] with focal loss [10]; F1 with 200 m geographic matching.
- Same optimizer, schedule, crops, data order and seed (0) for every cell. Strict FP32: 567.6 H100 GPU-hours.

Once-only held-out protocol

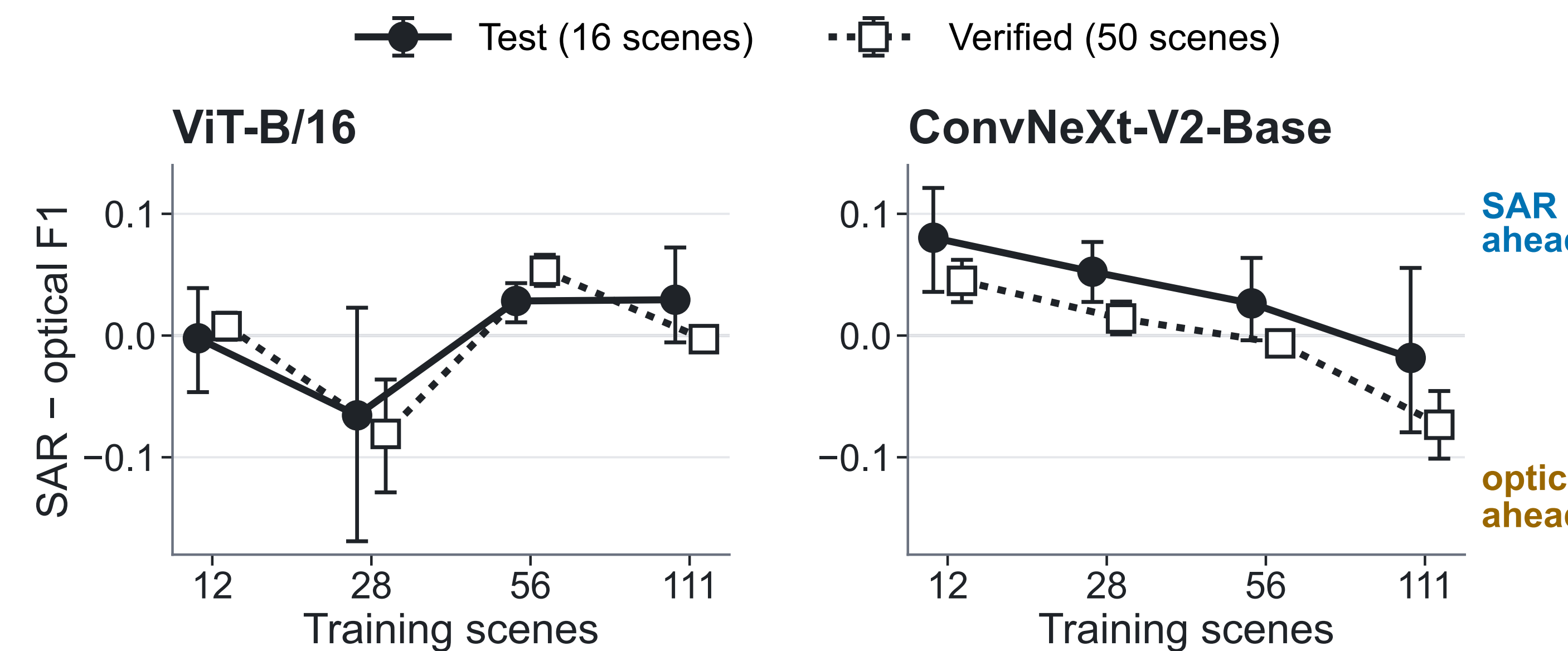
- Train 32 cells.** A sweep on 8 dev scenes picks each checkpoint and threshold.
- Freeze the cohort.** Hash-bind every marker and checkpoint.
- Score 16 test scenes once.** Same thresholds, **no retuning**.
- Open 50 verified scenes once.** Same thresholds, all 32 cells.

Pretraining helps most when labels are scarce



F1 with 200 m matching. Each of the 32 cells is scored once with the threshold it picked on 8 dev scenes. Top: 16 held-out test scenes. Bottom: 50 near-shore, human-verified scenes. Seed 0.

SAR vs. optical depends on backbone and budget



Bars: 95% intervals from 10,000 paired scene resamples. They cover scene sampling only: retraining 28 unchanged cells moved test F1 by up to 0.043.

The verified scenes expose a coastal label gap

0.4% → 31% 34 / 4,302 0.724

vessels within 2 km of shore: training labels → verified scenes best cell's predictions within 2 km of shore best cell's offshore verified F1 (0.574 overall)

Every cell reaches at most 0.021 near-shore recall and 0.228 dark-vessel recall, while precision stays at 0.639–0.909: the detectors learned to stay silent near shore.

Findings

12 / 12

pretrained-vs-random comparisons at 12 training scenes have a 95% interval above zero.

- 1 Pretraining pays off with few labels.** Gains at 12 scenes: +0.063 to +0.143 test F1, +0.048 to +0.111 verified F1.
- 2 SAR beats optical only for the CNN, and only with few labels:** +0.080 [+0.036, +0.121] test F1 at 12 scenes, but −0.073 [−0.101, −0.046] verified F1 at 111. ViT: no stable order.
- 3 The input adapter matters.** With a seeded 3-channel stem instead of band slicing, BigEarthNet-S2 beats random at every budget (+0.025 to +0.063 test F1).
- 4 The coastal gap is a label gap.** Only 0.4% of training vessels lie within 2 km of shore, against 31% of verified ones.

Limits

- One seed per cell; 8 dev scenes pick checkpoints and thresholds.
- Monotonicity gate failed: two SAR test curves drop by more than 0.02 as labels grow.
- Checkpoints differ in objective and corpus, not only domain.

Open questions & next steps

- How do multi-sensor foundation models such as CROMA [11] and DOFA [12] fare against single-source checkpoints?
- Does SAR pretraining help where optical cannot: near shore and for small dark vessels?
- Why does SARMAE peak at its first evaluation with 28 or fewer scenes?
- **Next:** near-shore training labels, a shoreline input channel, several seeds, and matched pretraining objectives across sensors.